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Reg. No: | | | | | | | | | | |

VELAGAPUDI RAMAKRISHNA

SIDDHARTHA ENGINEERING COLLEGE

(AUTONOMOUS)

IV/IV B.Tech. DEGREE EXAMINATION, NOVEMBER - 2023

Seventh Semester

INFORMATION TECHNOLOGY

20IT7301 DEEP LEARNING

Time: 3 hours ***Max. Marks: 70***

Part-A is compulsory

Answer One Question from each Unit of Part - B

Answer to any single question or its part shall be written at one place only

PART-A

10 x 1 = 10M

1. a. What is auto-association task in neural networks? **(CO1 K1)**
   b. Write the mathematical formula for summation function. **(CO1 K1)**
   c. Define over fitting. **(CO1 K1)**
   d. What is feature set? Give one example. **(CO2 K1)**
   e. Render simple architecture for auto encoder. **(CO2 K1)**
   f. Define max pooling. **(CO2 K1)**
   g. Render model for basic RNN. **(CO3 K1)**
   h. Specify layers in deep CNN **(CO3 K1)**
   i. Give two applications for Generative adversial neural network. **(CO4 K1)**
   j. Define conditional Generative adversial neural network. **(CO4 K1)**

PART-B

4 x 15 = 60M

UNIT-I

2. a. Describe architecture of feed forward neural networks. **(CO1 K2) 8M**

   b. Discuss the importance of training, validation and testing sets in modeling neural networks. **(CO1 K3) 7M**

(or)

3. a. Define linear neuron. List limitations of linear functions. **(CO1 K2) 7M**

   b. Explain and analyze delta rule with example. **(CO1 K3) 8M**

UNIT-II

4. a. Explain about CNN architecture. **(CO2 K3) 8M**

   b. Elaborate on CNN for NLP and Time Series. **(CO2 K3) 7M**

(or)

5. a. Explain about sparsity in auto encoders. **(CO2 K3) 7M**

   b. Describe how effective dimensionality reduction can be done using auto encoders. **(CO2 K3) 8M**

UNIT-III

6. a. Elaborate on sequence to sequence architectures in auto encoders. **(CO3 K3) 7M**

   b. Discuss on computational graphs using in sequence modeling. **(CO2 K3) 8M**

(or)

7. a. Describe about deep RNN for identifying hand written digits as case study. **(CO3 K3) 8M**

   b. Explain about Forward Propagation in Deep Networks and mathematical notations of hyper parameters. **(CO3 K3) 7M**

UNIT-IV

8. a. Discuss on various attention mechanisms applicable for recurrent models. **(CO4 K3) 8M**

   b. Illustrate steps required to generate image data using generative adversarial networks. **(CO4 K4) 7M**

(or)

9. a. Machine translation is challenging task in implementing real time use case. Justify on this statement with suitable examples. **(CO4 K4) 8M**

   b. List out various challenges of deep learning and neural networks. **(CO4 K3) 7M**

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